

enable the fluid to expand the top polymer layer to create physical buttons." This enables the buttons to bubble up when needed and flatten out when not in use. A controller is used to control the state of these buttons, while an application programming interface allows integrated software/application based control.

Build your own future phone

All of us have a vision of our future phone. How nice it would be if we could build it ourselves? That is what Dave Hakkens' Phonebloks project aims to achieve. This ongoing project, which started in 2012, aims to make a phone completely modular. It works towards a future where you can just assemble your own phone using building blocks. When you want to upgrade, simply change the blocks you wish to. If you have a great idea, build a new component and add it.

Google is into this great idea, too, with Project Ara. If you could get a basic phone for US\$ 50 and turn it into an advanced smartphone by adding all the great stuff you want, the company believes that that could lower the entry barriers in this industry. According to Google, it all starts with an endoskeleton, or endo, the structural frame and data backbone of the device. The user can populate the endo with modules, the building blocks that make up the vast majority of the phone's functionality and features.

The modules can be easily and safely inserted and removed at any time, even while the device is powered on. The modules also have user-replaceable covers or shells, which provide a creative canvas for users to make their phones look exactly as they wish.

Other modular phone efforts include Xiaomi Magic Cube, PuzzlePhone and FonKraft. If

What is a phone without energy

Designing futuristic phones is not just about form factor, displays and flexibility. Sustainability is all the more important. That makes designs like these worthy of consideration even if it would take years for their fruition.

Aeolus, designed by Cyrene Quiamco in 2010, can be charged using solar and wind power. The handset has a single-colour, energy-efficient liquid crystal display and is made of renewable materials. It has a small fan that can convert even mild breeze into electricity to charge the battery. It can also use kinetic energy from a moving vehicle to charge itself.

Since a few years, Kyocera and Sunpartner Technologies have been working on a transparent photovoltaic film that can be mounted above or below a phone's touchscreen to harness solar energy for its battery. The latest version of this Wysips, short for What You See Is Photovoltaic Surface, crystal panel is only 0.58mm thick and needs just three minutes of sun exposure to get a minute of smartphone talktime.

Eddie Goh's Eclipse Intuit is another interesting concept, in which parts of the phone are made from a chemically based thin solar skin that can charge the battery on sun exposure.

A Nokia concept phone designed by Daizi Zheng in 2010 runs on a bio-battery, which derives electricity from carbohydrates. Feed the phone some Coke and it will use the carbohydrates and sugar in the drink to power itself up. Once the battery dies only oxygen and water remain, making it completely bio-degradable. In his project outline, Zheng explains that on a single charge, bio-batteries have a potential life-span three to four times longer than conventional lithium batteries.



Nokia Green Phone (Image courtesy: Daizi Zheng)

you look at most of these modular phone projects, you will realise that these are not working against each other, but together in many ways—something that the future needs!

Modular phones would not only foster innovation but also reduce a lot of e-waste, helping a sustainable future. These would also do away

with the frustration that many of us face when we have just shelled out thousands of rupees to buy a phone, only to realise within a few months that the next generation is already here. If you love the idea of building your own phone, share your efforts and ideas to make these projects a success. **EFY**